L9: Comparison of SI, SIS, SIR Models Under Homogeneous Mixing

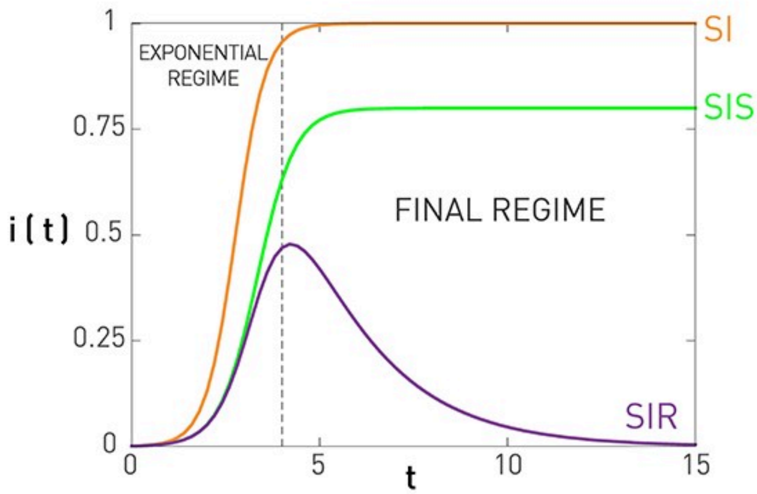

Figure 10.7 from Network Science by Albert-László Barabási

This figure summarizes the results for the SI, SIS, and SIR models, under the assumption of homogeneous mixing.

For the SI model there is no epidemic threshold and we always get an epidemic that infects the entire population.

For the SIS and SIR models we get an epidemic if the ratio $\frac{\bar{k}\,\beta}{\mu}$ is greater than the epidemic threshold, which is equal to one. In that case, both models predict an initial "exponential regime". The difference between the SIS and SIR models is that the former leads to an endemic state in which a fraction $1 - \frac{\mu}{\bar{k}\,\beta}$ of the population remains infected (if the epidemic threshold is exceeded).

There are more realistic models in the epidemiology literature, with additional compartmental states and parameters. A common such extension is to introduce an ***"Exposed"*** state E, between the S and I states, which models that individuals that are exposed to a pathogen stay dormant for some time period *(until they develop enough viral load)* before they become infectious. This leads to the SEIR model.

Another extension is to consider pathogens in which some infected individuals may acquire natural immunity while others may die. This requires to have two different Removed states, with different transition rates.